

# Experiment 'p\_6\_2\_b\_shift\_right\_array' Results

June 5, 2026

**Experiment outcome:** SUCCESS

**Bad responses:** 0

**Responses containing assume :** 0

**Responses containing expect :** 0

**Resolution attempts:** 1

**Hard fails (resolution):** 0

**Soft fails (resolution):** 0

**Verification attempts:** 1

## Problem Specification

**Problem name:** p\_6\_2\_b\_shift\_right\_array

**Natural language statement:** null

**Method signature:** p\_6\_2\_b\_shift\_right\_array(inputs: array<int>) returns (result: array<int>)

## Ensures

- `result.Length == inputs.Length`
- `result[0] == inputs[inputs.Length - 1]`
- `forall i :: 1 <= i < inputs.Length ==> result[i] == inputs[i - 1]`

## Requires

- `inputs.Length >= 1`

## GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

### Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_6_2_b_shift_right_array(inputs: array<int>) returns (result: array<int>)`

The method should respect the following contract:

requires `inputs.Length >= 1`, ensures `result.Length == inputs.Length`, ensures `result[0] ==`  
    ↪ `inputs[inputs.Length - 1]`, ensures forall `i :: 1 <= i < inputs.Length ==> result[`  
    ↪ `i] == inputs[i - 1]`

Produce and show only the Dafny body of this method, including the curly braces that  
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor  
    ↪ 'expect' anywhere in your code.

### GenAI → Program

**System fingerprint:** null

**ID:** chatcmpl-DnNikuwRpdYvP0OhCi0eiFi2jBfk1

**Model:** gpt-5.5-2026-04-23

**Created at:** 1780661626

**Finish reason:** stop

**Usage:** 338 tokens in, and 660 tokens out

```
{
  result := new int[inputs.Length];
  result[0] := inputs[inputs.Length - 1];

  var i := 1;
  while i < inputs.Length
    invariant 1 <= i <= inputs.Length
    invariant result.Length == inputs.Length
    invariant result[0] == inputs[inputs.Length - 1]
    invariant forall j :: 1 <= j < i ==> result[j] == inputs[j - 1]
  {
    result[i] := inputs[i - 1];
    i := i + 1;
  }
}
```

## Final Program

```

method p_6_2_b_shift_right_array(inputs: array<int>) returns (result: array<int>)
  requires inputs.Length >= 1
  ensures result.Length == inputs.Length
  ensures result[0] == inputs[inputs.Length - 1]
  ensures forall i :: 1 <= i < inputs.Length ==> result[i] == inputs[i - 1]
{
  result := new int[inputs.Length];
  result[0] := inputs[inputs.Length - 1];

  var i := 1;
  while i < inputs.Length
    invariant 1 <= i <= inputs.Length
    invariant result.Length == inputs.Length
    invariant result[0] == inputs[inputs.Length - 1]
    invariant forall j :: 1 <= j < i ==> result[j] == inputs[j - 1]
  {
    result[i] := inputs[i - 1];
    i := i + 1;
  }
}

```

## Total Token Usage

**Input tokens:** 338

**Output tokens:** 660

**Reasoning tokens:** 512

**Sum of ‘total tokens’:** 998

## Experiment Timings

**Overall Experiment** started at 1780661625727, ended at 1780661673312, lasting 47585ms (47.59 seconds)

**Iteration #1** started at 1780661625836, ended at 1780661673312, lasting 47476ms (47.48 seconds)